

COVID-19 Data Analysis

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Program: B.S CS (M2)

Semester: 6th

Course Title: Big Data Analysis

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Course Code: CS-506

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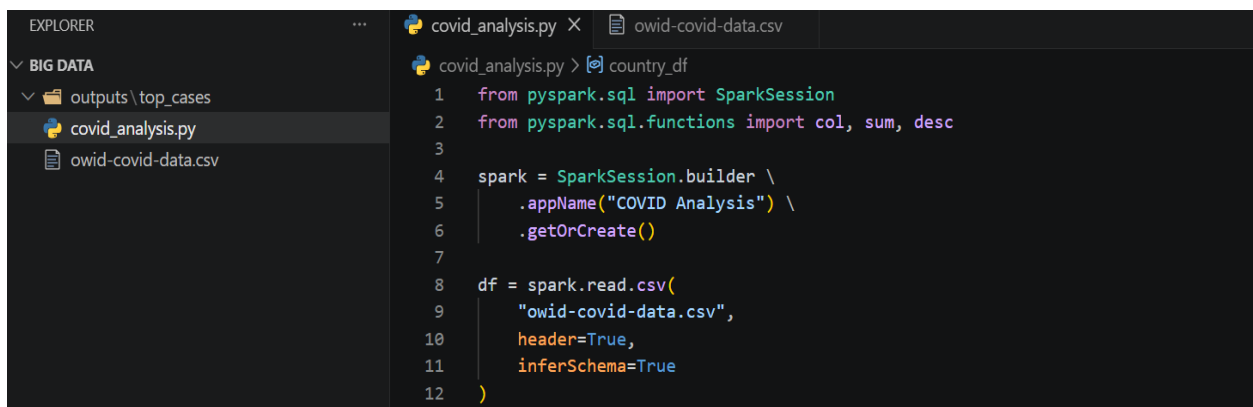
Tasks

Project Description

This mini project performs COVID-19 data analysis using PySpark and the Our World in Data (OWID) dataset. The project focuses on loading the dataset, filtering country-specific data, finding total cases and deaths, and sorting countries according to the highest number of COVID-19 cases.

The purpose of this project is to understand how big data tools like PySpark can process and analyze large datasets efficiently.

1. Load the COVID Dataset



The screenshot shows a code editor with two tabs: 'covid_analysis.py' and 'owid-covid-data.csv'. The 'covid_analysis.py' tab is active, displaying the following code:

```
1 from pyspark.sql import SparkSession
2 from pyspark.sql.functions import col, sum, desc
3
4 spark = SparkSession.builder \
5     .appName("COVID Analysis") \
6     .getOrCreate()
7
8 df = spark.read.csv(
9     "owid-covid-data.csv",
10    header=True,
11    inferSchema=True
12 )
```

This loads the dataset so we can perform different operations on it.

2. Filter Country-specific data

Filter COVID-19 records for a specific country such as China.



The screenshot shows a code editor with two tabs: 'covid_analysis.py' and 'owid-covid-data.csv'. The 'covid_analysis.py' tab is active, displaying the following code:

```
16
17 country_df = df.filter(col("location") == "China")
18
19 print("\nChina COVID Data")
20 country_df.select(
21     "date",
22     "total_cases",
23     "total_deaths"
24 ).show(20)
25
```

Output Table

China COVID Data

date	total_cases	total_deaths
2020-01-05	1	0
2020-01-06	1	0
2020-01-07	1	0
2020-01-08	1	0
2020-01-09	1	0
2020-01-10	1	0
2020-01-11	1	0
2020-01-12	45	1
2020-01-13	45	1
2020-01-14	45	1
2020-01-15	45	1
2020-01-16	45	1
2020-01-17	45	1
2020-01-18	45	1
2020-01-19	125	3
2020-01-20	125	3
2020-01-21	125	3
2020-01-22	125	3
2020-01-23	125	3
2020-01-24	125	3

only showing top 20 rows

3. Find total cases and deaths

Analyze the total COVID-19 cases and deaths.

This task displays the total number of COVID-19 cases and deaths recorded in China.

```
covid_analysis.py x owid-covid-data.csv
covid_analysis.py > ...
21
22
23
24
25
26 total_data = df.select(
27     sum("new_cases").alias("Total_Cases"),
28     sum("new_deaths").alias("Total_Deaths")
29 )
30
31 print("\nTotal cases and deaths")
32 (variable) total_data: DataFrame
33 total_data.show()
34
35
36
37
38
39
```

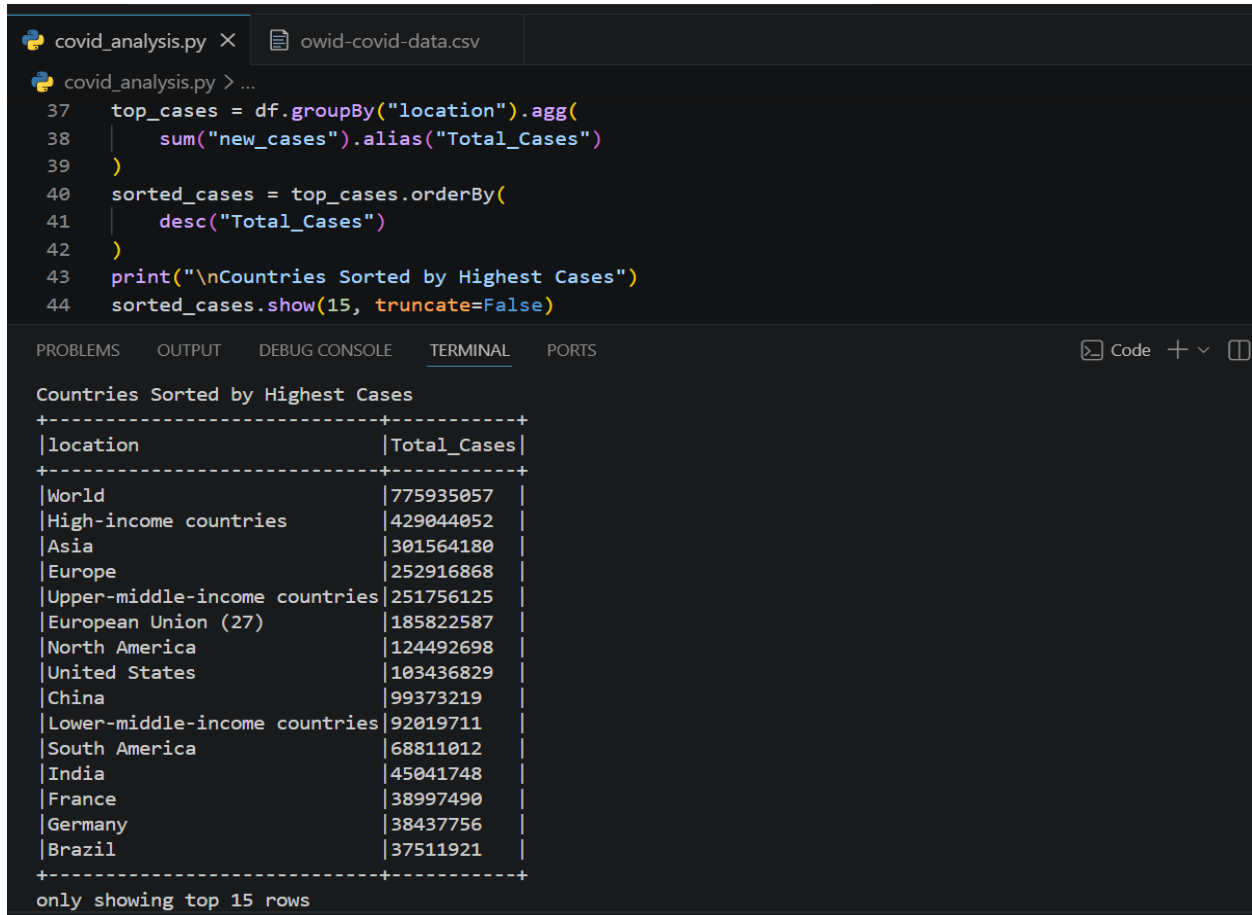
Terminal Output:

```
Total cases and deaths
+-----+-----+
|Total_Cases|Total_Deaths|
+-----+-----+
| 3288392333|    29503063|
+-----+-----+
```

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4. Sort by Highest cases

Sort countries according to the highest COVID-19 cases.



```
covid_analysis.py × | owid-covid-data.csv
covid_analysis.py > ...
37 top_cases = df.groupBy("location").agg(
38     sum("new_cases").alias("Total_Cases")
39 )
40 sorted_cases = top_cases.orderBy(
41     desc("Total_Cases")
42 )
43 print("\nCountries Sorted by Highest Cases")
44 sorted_cases.show(15, truncate=False)
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

Countries Sorted by Highest Cases

location	Total_Cases
World	775935057
High-income countries	429044052
Asia	301564180
Europe	252916868
Upper-middle-income countries	251756125
European Union (27)	185822587
North America	124492698
United States	103436829
China	99373219
Lower-middle-income countries	92019711
South America	68811012
India	45041748
France	38997490
Germany	38437756
Brazil	37511921

only showing top 15 rows

Insights

- In first step we loaded the data set named “owid-covid-data.csv” in our working environment to perform different operations on the data.
- In second step we filter country-specific data and the country we take is China and total cases and total deaths are shown. China was one of the first countries affected by COVID-19. Initial COVID-19 cases in China increased rapidly during the early outbreak period. The total number of deaths increased with the rise in total cases. PySpark efficiently processed and analyzed China’s large COVID-19 dataset.

- In third step we found Total cases and deaths overall. The purpose of this step was to observe the overall impact of COVID-19 on public health. By analyzing these values, we could compare the spread of the virus and the number of deaths caused by it.
- In forth step we apply Sorting by highest cases. The Sorting is an important data analysis technique because it helps users quickly identify trends and compare records. Through this step, we were able to analyze which countries experienced the largest outbreaks during the pandemic.